HBK/FJ-2008 COMPLETE DIE-SPRAY SYSTEM DESIGN

Application: 600-ton steel hot-forging press | Forging temperature: approximately 1200°C | Die: approximately Ø250–300 mm

Lubricant: HBK/FJ-2008 water-based polymer forging lubricant | Working dilution: 1 part lubricant : 7 parts water (user-supplied; verify with supplier TDS before production)

Objective: Compact adjustable pneumatic die-lubrication/cooling spray system, initially one nozzle and expandable to two nozzles.

> This is a practical starting specification, not a certified machine design. Final settings must be validated on the actual die, lubricant TDS/SDS, press cycle and plant compressed-air system.

1. FINAL SYSTEM CONFIGURATION

20–30 L stainless mixing tank → small pneumatic diaphragm pump → 80–150 mesh filter → liquid pressure regulator/gauge → 24 VDC normally-closed solenoid → CY WA-100-10 / WA-101, 1.0 mm air/liquid atomizing nozzle. The nozzle receives separate regulated atomizing air.

Start with one nozzle at approximately 200 mm from the die and approximately 0.5 s spray duration. If coverage is uneven, reposition or add a second nozzle rather than simply increasing lubricant quantity.

2. PROCESS DESIGN BASIS

Parameter	Design basis
Press	600-ton hot forging press
Material	Steel
Forging temperature	Approximately 1200°C
Die diameter / area	Approximately Ø250–300 mm
Lubricant	HBK/FJ-2008 water-based polymer forging lubricant
Dilution	1:7 lubricant:water — user-supplied working ratio; verify with supplier
Application	Die release, lubrication and cooling
Spray method	Pneumatic air/liquid atomization
Initial nozzle quantity	1
Expansion	2 nozzles if cavity geometry requires

3. SPRAY NOZZLE

Feature	Recommended specification
Model	CY WA-100-10 / WA-101 family
Orifice	1.0 mm
Type	Automatic pressure-feed air/liquid atomizing spray gun
Published liquid output	Approximately 190 ml/min; verify actual configuration
Published spray pattern	Approximately 250 mm width at 200 mm distance
Published air pressure	Approximately 3–4 bar
Air consumption	Approximately 270 L/min
Connections	1/4 inch air and fluid
Initial distance	Approximately 200 mm
Pattern	Wide/flat, uniform thin film
Initial spray time	0.3–0.5 s; start near 0.5 s

Supplier: CY WA-101 manufacturer/product page
Purchase listing: Made-in-China listing

> Current price must be confirmed directly with the supplier; do not rely on an old/sample price.

4. PNEUMATIC DIAPHRAGM PUMP

Feature	Recommended specification
Type	Air-operated double-diaphragm (AODD)
Size	Compact 1/4-inch class for one nozzle
Wetted material	SS316 or PP; PTFE/EPDM only after compatibility check
Adjustable flow	Approximately 0.2–5 L/min target range
Maximum pressure	At least 0.5 MPa with downstream regulation
Air supply	Clean, dry compressed air
Typical pump air pressure	3–6 bar, according to pump
Purpose	Stable liquid supply for short spray pulses
Selection priority	Low-flow controllability rather than very high maximum flow

Example industrial family: Ingersoll Rand PD01P / 1/4-inch AODD class. Chinese equivalents are available; confirm wetted materials, low-flow behavior and pressure rating before purchase.

5. MIXING TANK + AGITATION

Feature	Recommended specification
Capacity	20–30 L for initial one-nozzle system
Material	SS304; SS316 preferred if required
Lid	Closed lid with filling opening
Agitation	Continuous mechanical agitation or pump recirculation
Agitator speed	Approximately 100–300 rpm starting range
Baffles	Recommended
Bottom	Sloped/rounded for drainage
Drain	Bottom drain valve
Level	Level tube or sensor
Filtration	80–150 mesh before nozzle
Cleaning	Easy access and flush/drain

Maintain a uniform mixture. Avoid excessive agitation that creates unnecessary air entrainment.

6. LIQUID CIRCUIT

`20–30 L TANK → AODD PUMP → 80–150 MESH FILTER → LIQUID REGULATOR → NC 24 VDC SOLENOID → NOZZLE → DIE`

Component	Specification
Liquid hose	1/4 inch, chemically compatible
Filter	80–150 mesh, cleanable
Liquid regulator	Adjustable, approximately 0–4 bar
Liquid gauge	0–4 bar
Solenoid	NC, 24 VDC, 1/4 inch, chemically compatible wetted parts
Check valve	Recommended
Ball valves	Isolation/drain/maintenance
Fittings	SS304/316 preferred

7. COMPRESSED-AIR CIRCUIT

`PLANT AIR → FILTER/WATER SEPARATOR → REGULATOR → GAUGE → NOZZLE AIR PORT`

Feature	Recommended specification
Supply	Clean, dry plant compressed air
Regulator	0–6 or 0–10 bar
Initial nozzle air	Approximately 3–4 bar for WA-100-10 published range
Air hose	1/4 inch; verify pressure drop
Gauge	0–6 bar near nozzle
Water separator	Recommended

Feature	Recommended specification
Isolation valve	Recommended

Use the nozzle manufacturer's air requirement as the controlling specification for the selected nozzle.

8. CONTROL — SOLENOID + TIMER / PLC

Feature	Recommended specification
Control voltage	24 VDC preferred
Liquid valve	Normally closed (NC)
Timer range	0.1–2.0 s adjustable
Initial spray time	Approximately 0.5 s
Manual mode	Required for setup
Automatic mode	Press-cycle triggered
Monitoring	Liquid and air pressure gauges
Low level	Recommended
Emergency stop	Required when integrated with press
Fault indication	Recommended

9. PRESS SYNCHRONIZATION

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PRESS SAFE SPRAY POSITION → PRESS SIGNAL → SPRAY ON → 0.3–0.5 s → SPRAY OFF → FORGING CONTINUES
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Feature	Recommendation
Press interface	Dry contact or 24 VDC signal from press PLC
Spray delay	Adjustable, approximately 0–2 s
Spray duration	Adjustable, approximately 0.1–2 s
Manual spray	Pushbutton for setup
Interlock	Spray only at correct press/die position
Emergency stop	Immediately removes spray command
Cycle counter	Useful

10. INITIAL PROCESS SETTINGS

Parameter	Initial trial	Comment
Nozzle	WA-100-10, 1.0 mm	One nozzle first
Distance	≈200 mm	Tune to cavity
Liquid pressure	≈0.15–0.25 MPa	Starting range only
Atomizing air	≈3–4 bar	WA-100-10 published range
Spray time	≈0.5 s	Adjust 0.3–0.5 s
Published flow	≈190 ml/min	Verify actual flow
Liquid per 0.5 s	≈1.6 ml	Calculated, not production dose
Coverage	Uniform thin film	Avoid flooding

The liquid-pressure range is a starting process range, not a validated HBK/FJ-2008 production specification. Final settings must be established experimentally.

11. BILL OF MATERIALS

Item	Specification	Qty.
Spray nozzle	CY WA-100-10 / WA-101, 1.0 mm	1
AODD pump	1/4-inch class, PTFE-compatible, adjustable low flow	1
Mixing tank	SS304/316, 20–30 L, lid + drain	1
Agitator	100–300 rpm starting range	1
Liquid filter	80–150 mesh, cleanable	1
Liquid regulator	0–4 bar adjustable	1

Item	Specification	Qty.
Liquid gauge	0–4 bar	1
Liquid solenoid	1/4-inch, NC, 24 VDC, compatible wetted parts	1
Air filter/regulator	1/4-inch, water separator, 0–6/10 bar	1
Air gauge	0–6 bar	1
Timer/PLC	24 VDC, 0.1–2 s timing	1
Check valve	1/4-inch, liquid-compatible	1
Ball valves	1/4-inch / suitable tank drain	2–3
Liquid hose	1/4-inch chemical-compatible	3–5 m
Air hose	1/4-inch	3–5 m
SS fittings	SS304/316, 1/4-inch as required	Set
Pressure relief/bypass	Set below system maximum	1
Low-level sensor	24 VDC, tank-compatible	Optional

12. SUPPLIER / PURCHASE LINKS

Source	Link
CY WA-101	https://cycospray.com/product/high-quality-hot-sale-pressure-feed-wa-101-wa200-new-pneumatic-automatic-paint-spray-gun/
CY Made-in-China	https://cycospray.en.made-in-china.com/product/rwMtmUibnohF/China-Wa-101-Automatic-Spray-Tools-Assembly-Line-Special-Spray-Gun-for-Reciprocating-Machine.html
Shanghai Xinhou spray nozzle	https://shxinhou.en.made-in-china.com/product/INkmogQxgZcP/China-Low-Pressure-Atomizing-St-Dk-Automatic-Spray-Mold-Release-Agent-Spray-Gun-Nozzle.html
BETE lubrication	https://bete.com/application/lubrication/
BETE BJ series	https://bete.com/product/bj/
MAFA TMKDUO	https://www.mafa-shop.com/en/Releasing-Agent-Spray-Gun-TMKDUO/180204-MAIN
AYANK automatic die lubrication	https://www.ayankforging.com/automated-mold-lubrication-and-cooling-equipment.html
Jinan Linteng	https://jnlinteng.en.made-in-china.com/product/NFyTUMXvSOYm/China-Automation-Graphite-Spray-Machine-for-Forging-Die-Lubrication.html
Nanjing Descaling	https://www.descalingtech.com/automatic-graphite-spray-station-for-forging-line_p83.html
Xuzhou Dayi	https://www.dayiforgingtech.cn/solution/83.html
Steel Plant Specialities	https://steelplantspecialities.com/die-lubricant-spraying-systems/
HBK/FJ-2008 manufacturer	https://www.qdxyfsm.com/

13. COMMISSIONING PROCEDURE

1. Verify the HBK/FJ-2008 dilution from the supplier TDS. The 1:7 ratio in this document is user-supplied.
2. Fill the tank with clean/filtered water and add lubricant.
3. Run continuous agitation/recirculation and confirm a uniform mixture.
4. Check all liquid and air connections for leaks.
5. Start the pump at low pressure and purge air from the liquid line.
6. Confirm the filter is clean.
7. Set nozzle air near 3–4 bar for the initial WA-100-10 trial.
8. Set liquid pressure conservatively and increase only until the pattern is stable.
9. Position nozzle about 200 mm from the die.
10. Test in manual mode first.
11. Start near 0.5 s spray time and inspect the film.
12. Run hot-forging trials and inspect release, die wetting, cooling, surface condition and lubricant consumption.

13. Only after stable manual operation connect the automatic press-cycle trigger.

14. SAFETY AND ENGINEERING NOTES

• Hot surface: protect hoses, valves and controls from radiant heat, scale and direct contact with the hot die.

• Chemical compatibility: verify diaphragm, seals, hoses, filter and valve wetted materials against the HBK/FJ-2008 TDS/SDS.

• Pressure: protect the circuit with suitable pressure relief/bypass; never exceed the weakest component rating.

• Air: use clean, dry compressed air.

• Electrical: use 24 VDC controls and proper press interlocks.

• Filter maintenance: nozzle blockage can change spray pattern and dose.

• Two-nozzle expansion: add a second nozzle if geometry requires coverage; avoid solving poor coverage only by increasing spray quantity.

• Production validation: final pressure, time, distance and concentration must be validated on the actual die and cycle.

15. FINAL RECOMMENDATION

Build the first system around one CY WA-100-10 / WA-101, 1.0 mm nozzle, a 20–30 L SS tank, small 1/4-inch AODD pump, continuous agitation/recirculation, 80–150 mesh filter, liquid regulator, 24 VDC NC solenoid, and a 24 VDC timer/PLC. Start at approximately 200 mm nozzle distance, 3–4 bar atomizing air, and approximately 0.5 s spray time.

> Do not finalize the pump, solenoid, seal materials or production dose until HBK/FJ-2008 compatibility is confirmed. The key acceptance criterion is a uniform thin film with reliable release and controlled lubricant consumption.

Product specifications, availability and prices can change. Confirm current specifications, compatibility, price and delivery directly with the supplier before ordering.